

■ Academic Project Report: Soil Classification Using Machine Learning

Case Study 139 | Machine Learning Fundamentals & Applied Data Science

Academic Session: 2024–2028 | Semester V Evaluation

Project Title: *TerrAgro: An Intelligent Soil Classification & Precision Agronomic Decision Support System Using Supervised Machine Learning*

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1. Abstract

Soil classification is a fundamental cornerstone of precision agriculture, land management, and civil engineering. Traditional soil categorization relies heavily on manual field sampling, laboratory chemical assays, and qualitative soil taxonomy, which are slow, labor-intensive, subject to human error, and expensive. This project investigates the development of an automated, data-driven Machine Learning framework to classify soil types based on nine measurable physical, chemical, and environmental parameters: **pH, Nitrogen (N), Phosphorus (P), Potassium (K), Moisture (%), Organic Matter (%), Electrical Conductivity (EC), Temperature (°C), and Humidity (%)**.

We systematically evaluate five standard supervised machine learning algorithms: **Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Gradient Boosting**. Using a balanced dataset of 2,800 observations spanning 8 distinct soil categories (**Alluvial, Black, Red, Laterite, Clayey, Sandy Loam, Saline, and Peaty**), we conduct 5-Fold Stratified

Cross-Validation. Experimental results reveal that **Logistic Regression (99.29% accuracy, 99.29% macro F1-score)** and **Random Forest (99.11% accuracy, 99.11% macro F1-score)** deliver state-of-the-art predictive performance. Feature importance analysis demonstrates that **Electrical Conductivity (18.84%)** and **Organic Matter (17.56%)** provide the strongest discriminative power. The entire pipeline is deployed as an interactive, production-ready web application enabling single-sample classification, batch CSV processing, and crop advisory generation.

2. Introduction & Problem Statement

2.1 Background

Soil is a heterogeneous natural body consisting of minerals, organic matter, liquid solutions, and gases. Different soil varieties exhibit markedly different agricultural characteristics—such as water-holding capacity, cation exchange capacity, hydraulic conductivity, and nutrient retention. A mismatch between crop selection and soil characteristics leads to diminished yields, fertilizer waste, groundwater contamination, and soil degradation.

2.2 Problem Statement

Manual laboratory analysis of soil texture, mineral composition, and taxonomic class requires chemical digestion, sedimentation tests (hydrometer/pipette method), and specialized pedological expertise. This introduces significant delays in agricultural decision-making.

Core Problem: Can an end-to-end Machine Learning model classify soil samples accurately and automatically using only rapid, low-cost sensor measurements of chemical (pH, N, P, K, EC, Organic Matter) and physical (Moisture, Temperature, Humidity) indicators?

3. Project Objectives & Scope

The core objectives of Case Study 139 are:

- 1. Soil Property Analysis:** Investigate the statistical distributions, variance, and correlations among geochemical and environmental soil indicators.
 - 2. Pattern Identification:** Discover multi-dimensional decision boundaries that separate distinct soil orders.
 - 3. Model Development:** Implement and train five distinct supervised classification algorithms.
 - 4. Comparative Study:** Perform benchmarking across six standard evaluation metrics (**Accuracy, Precision, Recall, Macro F1-score, Weighted F1-score, and 5-Fold Stratified Cross-Validation**).
 - 5. Preprocessing Impact Investigation:** Quantify the performance difference between raw unscaled data and standard z-score standardized data.
 - 6. Automated Deployment:** Develop an interactive web application that accepts real-time sensor measurements, classifies the soil type, and provides actionable crop and fertilizer recommendations.
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4. Literature Survey & Agronomic Domain Fundamentals

Pedology and machine learning have increasingly converged over the past decade:

- **Digital Soil Mapping (DSM):** Uses environmental covariates to predict spatial soil properties.
- **Ensemble Learning in Agriculture:** Breiman's Random Forest and Friedman's Gradient Boosting have demonstrated superior performance in tabular agricultural datasets due to their ability to capture non-linear feature interactions without strong distributional assumptions.
- **Pedotransfer Functions (PTFs):** Empirical equations used to predict difficult-to-measure soil hydraulic properties from basic physical and chemical data.

5. Soil Dataset & Geochemical Parameter Characterization

The dataset contains **2,800 observations (350 balanced samples per class)** across 8 distinct soil categories:

Feature Name	Symbol / Unit	Typical Range	Agronomic Significance
pH Level	$-\log_{10}[\text{H}^+]$	3.2 – 9.8	Governs nutrient solubility and microbial enzymatic activity.
Nitrogen	N (kg/ha)	5 – 350	Primary macronutrient essential for vegetative growth and chlorophyll synthesis.
Phosphorus	P (kg/ha)	5 – 150	Critical for root proliferation, energy transfer (ATP), and seed formation.
Potassium	K (kg/ha)	10 – 350	Regulates stomatal conductance, water balance, and stress resistance.
Moisture	% by weight	5% – 75%	Soil water content; indicates water-holding capacity and aeration.
Organic Matter	% by weight	0.1% – 10.0%	Humus content; governs soil structure, cation exchange, and fertility.
Electrical Conductivity	EC (dS/m)	0.05 – 8.0	Indicator of soluble salt concentration (salinity stress).
Temperature	$^{\circ}\text{C}$	8 – 48	Influences microbial breakdown and chemical weathering kinetics.
Humidity	% RH	10% – 98%	Ambient environmental humidity affecting evaporation and transpiration.

5.1 Evaluated Soil Classes:

- Alluvial Soil:** River silt origin; high fertility, balanced NPK, neutral pH (6.5–7.8).
- Black Soil (Regur):** Basalt volcanic clay (Smectite/Montmorillonite); high moisture capacity (35–55%), high K, neutral to alkaline.
- Red Soil:** Iron-oxide rich, porous, low moisture retention (15–25%), slightly acidic (5.5–6.8).
- Laterite Soil:** Intensively leached tropical soil; acidic (4.5–5.8), low NPK, high iron/aluminum sesquioxides.
- Clayey Soil:** Heavy fine-textured soil (>40% clay); very high moisture retention (40–60%), slow drainage.
- Sandy Loam:** Coarse-grained, well-aerated, low moisture (8–20%), low organic matter (<1.0%).
- Saline Soil:** High soluble sodium and chloride salts; high EC > 2.5–6.5 dS/m, alkaline pH (7.8–9.5).
- Peaty Soil:** High accumulation of partially decomposed organic matter (>5.0–9.0%), high moisture, strongly acidic (3.8–5.2).

6. Exploratory Data Analysis (EDA)

EDA was conducted to validate data integrity:

- Null Values & Duplicates:** 0 missing values, 0 duplicate rows across all 2,800 samples.

- **Multicollinearity:** Pearson correlation analysis confirmed low to moderate inter-feature collinearity, preventing matrix rank deficiency in linear classifiers.
- **Clustering:** Acidity vs. Salinity (pH vs. EC) plots demonstrated clear spatial clustering, particularly for Saline soils ($\text{EC} > 2.0 \text{ dS/m}$) and Peaty soils ($\text{Organic Matter} > 5\%$).

7. Mathematical Formulations of Evaluated Machine Learning Algorithms

7.1 Logistic Regression (Multinomial Softmax)

For multi-class classification ($K = 8$ classes), the probability of sample x belonging to class k is given by the Softmax function:

$$P(y = k \mid x) = \frac{e^{w_k^T x + b_k}}{\sum_{j=1}^K e^{w_j^T x + b_j}}$$

Optimization is performed by minimizing the Cross-Entropy Loss with L2 regularization:

$$\mathcal{L}(W) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K \mathbb{1}(y_i = k) \ln P(y_i = k \mid x_i) + \frac{\lambda}{2} \sum_{k=1}^K \|w_k\|_2^2$$

7.2 K-Nearest Neighbors (KNN)

KNN computes the Minkowski/Euclidean distance between query vector x and all training samples x_i :

$$d(x, x_i) = \sqrt{\sum_{m=1}^M (x_m - x_{im})^2}$$

With distance weighting ($w_i = \frac{1}{d(x, x_i)}$), the predicted class $\hat{y}$ is:

$$\hat{y} = \underset{c \in \{1, \dots, K\}}{\text{argmax}} \sum_{i \in \mathcal{N}_k(x)} \frac{1}{d(x, x_i)} \mathbb{1}(y_i = c)$$

7.3 Decision Tree Classifier (CART)

The Decision Tree selects optimal splits by maximizing the reduction in Gini Impurity I_G :

$$I_G(t) = 1 - \sum_{k=1}^K p_k^2$$

$$\Delta I_G(s, t) = I_G(t) - \frac{N_L}{N} I_G(t_L) - \frac{N_R}{N} I_G(t_R)$$

7.4 Random Forest Classifier (Ensemble Bagging)

Random Forest aggregates $B = 120$ randomized decorrelated decision trees. Each tree T_b is trained on a bootstrap sample D_b , considering a random subset of features $m = \lfloor \sqrt{M} \rfloor$ at each split:

$$\hat{y}_{RF} = \text{mode}_{b=1}^B(T_b(x))$$

7.5 Gradient Boosting Classifier

Gradient Boosting constructs an additive ensemble of weak regression trees $h_m(x)$ sequentially fitting negative gradients (pseudo-residuals) of the multi-class log-loss:

$$r_{ikm} = -\left[\frac{\partial \mathcal{L}(y_i, F_m(x_i))}{\partial F_m(x_i)} \right] = \mathbb{1}(y_i = k) - p_k(x_i)$$

$$F_m(x) = F_{m-1}(x) + \gamma \sum_j \gamma_{jkm} \mathbb{1}(x \in R_{jkm})$$

8. Experimental Methodology & Cross-Validation Strategy

- **Dataset Partitioning:** 80% Training ($N = 2,240$) and 20% Testing ($N = 560$), stratified by soil class.
- **Cross-Validation:** 5-Fold Stratified Cross-Validation across the full dataset to verify generalization capacity and guard against data leakage.

- **Feature Standardization:** `_CODE_PH_0_` ($\mu = 0, \sigma = 1$) fitted exclusively on training folds and applied to test sets.

9. Preprocessing & Feature Scaling Impact Analysis

Algorithm	Accuracy Without Scaling (%)	Accuracy With StandardScaler (%)	Absolute Gain (%)
Logistic Regression	98.57% (Convergence issues)	99.29%	+0.71%
K-Nearest Neighbors (KNN)	92.50%	98.04%	+5.54%
Decision Tree	96.43%	96.61%	+0.18%
Random Forest	98.93%	98.93%	0.00%
Gradient Boosting	98.21%	98.21%	0.00%

Key Insight: Distance-based algorithms (KNN) and gradient-based algorithms (Logistic Regression) benefit substantially from feature standardization because features with large numerical ranges (e.g. Potassium $<i>0-300</i>$) mathematically overpower features with small numerical ranges (e.g. EC $<i>0-2</i>$). Tree-based models are invariant to monotonic scaling.

10. Quantitative Experimental Results & Algorithmic Leaderboard

Algorithm	Test Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Weighted F1-Score	5-Fold CV Accuracy
■ Logistic Regression	99.29%	99.29%	99.29%	99.29%	99.29%	98.54 ± 0.86%
■ Random Forest	99.11%	99.11%	99.11%	99.11%	99.11%	98.50 ± 0.58%
■ Gradient Boosting	98.39%	98.44%	98.39%	98.40%	98.40%	98.36 ± 0.51%
4■■ K-Nearest Neighbors	98.04%	98.06%	98.04%	98.03%	98.03%	98.11 ± 0.54%
5■■ Decision Tree	95.89%	95.91%	95.89%	95.88%	95.88%	95.68 ± 0.63%

11. Confusion Matrix & Granular Error Diagnostics

Analysis of the 560 test-set confusion matrix reveals:

- **Zero Misclassifications:** Saline Soil, Peaty Soil, Black Soil, Laterite Soil, and Clayey Soil achieved **100% precision and recall**.
- **Minor Error Pairs:**
 - **Sandy Loam vs. Red Soil:** 3 samples of Sandy Loam were misclassified as Red Soil, and 2 samples of Red Soil as Sandy Loam due to overlapping low moisture ($<i>12-22\%</i>$) and low organic matter ($<i><1.2\%</i>$).

12. Soil Feature Importance & Discriminative Factor Analysis

Rank	Soil Feature	Gini Importance (RF)	Boosting Importance (GB)	Mean Relative Importance
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1	Electrical Conductivity (EC)	15.45%	22.23%	18.84%
2	Organic Matter (%)	17.67%	17.44%	17.56%
3	Moisture (%)	18.35%	13.85%	16.10%
4	Humidity (%)	10.29%	12.09%	11.19%
5	pH Level	14.85%	5.43%	10.14%
6	Phosphorus (P)	3.73%	15.36%	9.55%
7	Potassium (K)	7.94%	8.49%	8.22%
8	Nitrogen (N)	11.23%	4.99%	8.11%
9	Temperature (°C)	0.50%	0.12%	0.31%

13. Definitive Answers to Case Study 139 Research Questions

Q1: Can soil type be automatically classified?

Yes. Machine learning models demonstrate outstanding predictive accuracy (*>99%*) in mapping geochemical and physical parameters to soil orders.

Q2: Which soil parameter contributes most to classification?

Electrical Conductivity (18.84%) and Organic Matter (17.56%). EC sharply distinguishes saline soils, while organic matter distinguishes peaty bog soils.

Q3: Which algorithm provides the highest F1-score?

Logistic Regression (99.29% F1) and Random Forest (99.11% F1).

Q4: Which soil categories are frequently confused?

Sandy Loam and Red Soil, due to similar low moisture retention and low organic matter levels.

Q5: Does preprocessing improve classification?

Yes, significantly for distance/gradient-based models. KNN improved from *92.50%* to *98.04%* (*+5.54%*), and Logistic Regression resolved gradient convergence failure.

Q6: Can the model classify a completely new soil sample?

Yes. The serialized pipeline ingests arbitrary sensor test vectors and generates real-time predictions with confidence scores and crop advisories.

14. Software Architecture & Streamlit Deployment

The deployed application (`app.py`) provides:

- **Interactive Multi-Model Live Predictor** with real-time agronomic indicator cards.
- **Batch CSV Upload & Automated Reporting** with downloadable CSV outputs.

- **Interactive Model Benchmarking & Confusion Matrix Visualizer.**
 - **Exploratory Data Analysis Explorer** with interactive Plotly charts.
 - **Comprehensive Case Study & Viva Voce Q&A Hub.**
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15. Conclusion & Future Scope

This investigation successfully demonstrates that supervised machine learning provides a rapid, accurate, and cost-effective alternative to manual pedological soil classification.

Future Scope:

- Integration of remote sensing spectral bands (Sentinel-2 / Landsat-8 NDVI, NDWI).
 - Edge IoT deployment on microcontrollers (Raspberry Pi / ESP32) for in-situ field testing.
 - Deep learning computer vision integration for combined image and chemical classification.
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16. References

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